



Python for Fixed Income

COURSE 5 · ADVANCED MACHINE LEARNING

Bagging and Out-of-Bag Scores

WEEK 1 · DAY 2

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Bagging, and a score that costs no rows

1. Draw

A bootstrap sample of the training rows

2. Fit

One full tree on it

3. Average

Repeat, and average the trees' outputs

Today's two questions

How many trees before more stop helping?

Can the rows each tree never saw score the whole model, and what does scikit-learn's ready-made score measure?

Bagging helps unstable models. A full tree is as unstable as models come.

A bagged model is the average of its trees

0.6015 to 0.6219

single full trees

0.7243

their average, by hand

0.7320

BaggingClassifier, 10 trees

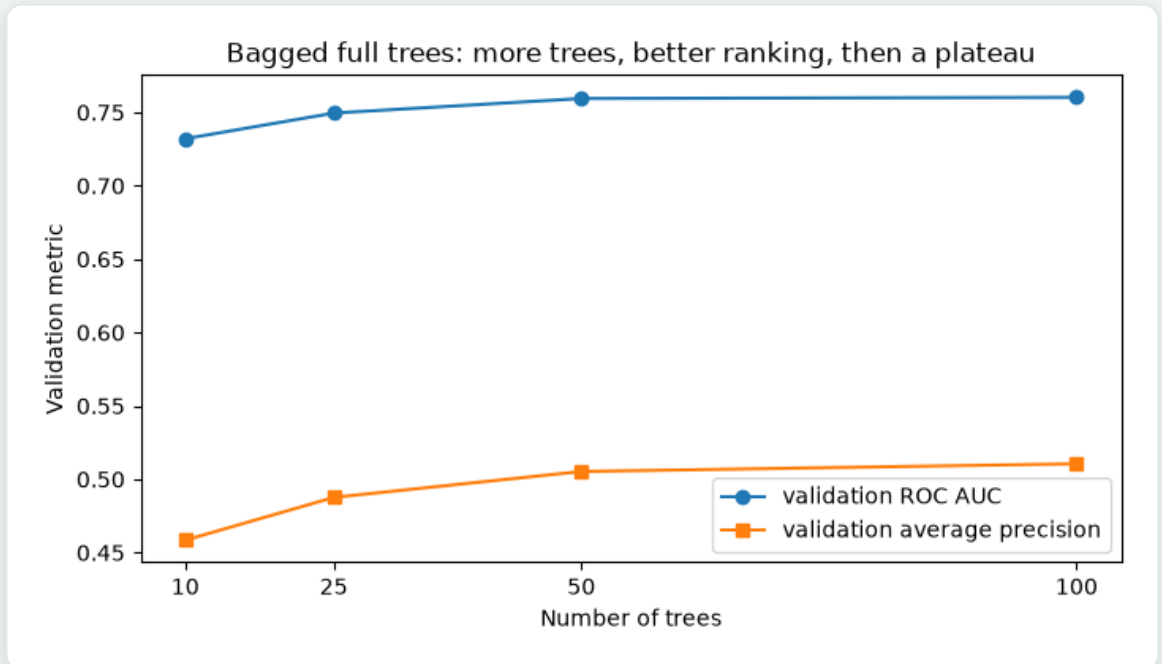
Validation ROC AUC; scikit-learn draws its own ten samples from the seed

```
by_hand = ensemblekit.bagged_proba(bagging_10, X_valid)
np.abs(by_hand - proba_10).max() < 1e-12    # True on every validation account
```

Each tree sees 18,000 draws holding 11,321 to 11,423 distinct rows. The output is their plain average.

More trees, better ranking, then a plateau

Trees	Validation ROC AUC	Average precision
10	0.7320	0.4584
25	0.7495	0.4876
50	0.7594	0.5053
100	0.7602	0.5105



10 to 25 trees: +0.0175. 50 to 100 trees: +0.0008. Logistic regression: 0.7190.

Every training row, scored by trees that never saw it

Each tree's sample leaves a row out with chance about 36.8 percent. Average only those trees' outputs for that row: an honest output, from one fit, with no rows held back.

36.8

trees that never saw a row, on average

17 to 56

fewest to most

True

same trees as without it

```
BaggingClassifier(DecisionTreeClassifier(random_state=SEED), n_estimators=100, oob_score=True, ...)
```

`oob_decision_function_` holds each row's out-of-bag output. 11 tests pass at the end of today.

oob_score_ is accuracy, not ROC AUC

0.8109

oob_score_ (accuracy)

0.7788

flag no account

0.7502

out-of-bag ROC AUC

0.7602

validation ROC AUC

A high floor

Most accounts do not default, so saying "no default" is right 0.7788 of the time. oob_score_ is only a little above it.

Why it leans low

Each row's out-of-bag output averages about 37 trees, not 100. The 25-tree model scored 0.7495 on validation.

Compare models by `ensembleskit.oob_auc`, never by `oob_score_` beside a ROC AUC.

Where out-of-bag helps: a small file

The first 2,000 training rows only, 415 defaults. A 100-tree bagged model, scored two ways.

0.7325

out of bag, one fit

0.7325

five-fold mean, five fits

0.6994 to 0.7630

the five folds

ROC AUC. Fold standard deviation 0.0225; the one-fit model on the validation rows: 0.7366.

On a bagged model or a forest, the out-of-bag ROC AUC is a fair stand-in for cross-validation when rows are few.

Bagging a stable model changes almost nothing

	Logistic regression	Full trees
Validation ROC AUC, one model	0.7190	0.6015 to 0.6219
Validation ROC AUC, bagged	0.7195 (25 copies)	0.7602 (100 trees)
Average spread of members on an account	0.0121	0.3365

Members that barely disagree leave nothing to average away. Bagging helps unstable models.

What bagging cannot fix

First question of the 100 trees

PAY_0 <= 1.5: 99 trees

PAY_0 <= 0.5: 1 tree

0.2581

average correlation of two trees' outputs (4,950 pairs)

The part every tree shares cannot average away. Day 3: let each split see only a few columns, so the trees disagree.

Excel side by side

Excel side by side (tested in Excel)

`=AVERAGE(B2:K2)`, filled down

`=SUMPRODUCT(--((B2:B18001>0.5)=(A2:A18001=1)))/ROWS(A2:A18001)`

`>=0.5` in place of `>0.5`

`=1-AVERAGE(A2:A18001)`

`=SORT(UNIQUE(B2:B18001),,-1)`, rates by `MAP(LAMBDA)`, trapezoid

equals `predict_proba` on all 6,000 accounts

0.8109: `oob_score_` exactly

0.8099: ties at 0.5 count as no default

0.7788, the all-zero baseline

0.7502, within 1e-6

No bagging and no tree in Excel. 10 of 11 near-equal out-of-bag pairs compared as equal in Excel, hence 1e-6.

Save the day: git diff on the log

1. Code

mlkit.py, ensemblekit and its tests

2. Log

results.csv with the 10-tree model

3. Diff

two rows added: + marks them

4. Commit

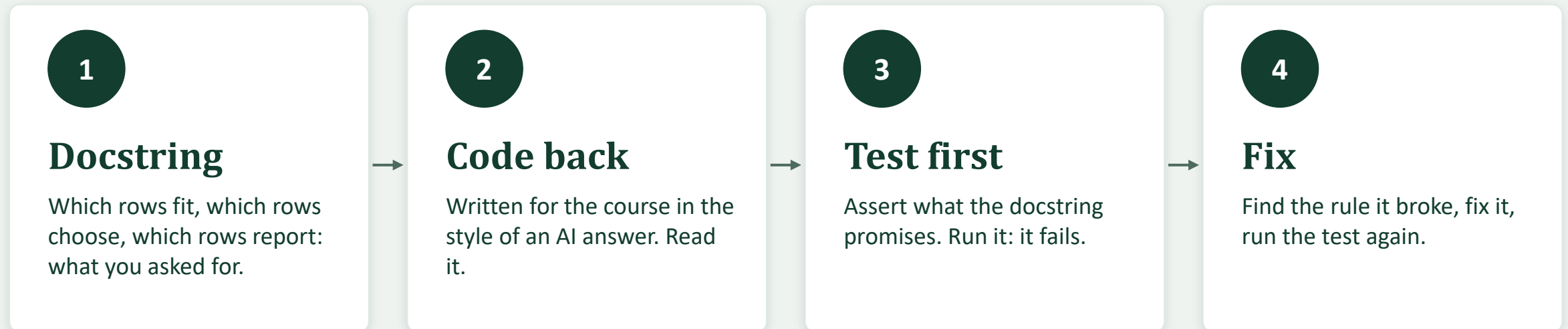
the log, three commits in all

```
bagging_10_trees,validation,0.7320,0.4584,0.5000,0.7943,0.5471,0.4069,0.4667
+bagging_100_trees,validation,0.7602,0.5105,0.5000,0.8143,0.6357,0.3760,0.4725
+bagged_logistic_25,validation,0.7195,0.4952,0.5000,0.8117,0.7223,0.2411,0.3616
```

Four decimals and LF endings: the unchanged row shows no mark, so a diff shows only real changes.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.



Today: `oob_score_`, an accuracy, is returned as the out-of-bag ROC AUC.

Code that runs is not the same as code that is right.

What you can do now

- BaggingClassifier is the plain average of its trees
- More trees help, then the curve flattens
- `oob_score_` is accuracy; `oob_auc` is the ranking number
- Out of bag stands in for cross-validation on few rows
- Bagging helps unstable models, not a logistic regression

Week 1, Day 3 Random forests: trees that disagree on purpose